

Roll No.

TBT-801

**B. TECH. (BIOTECHNOLOGY) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022**

OMICS TECHNOLOGIES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you understand by next generation sequencing ? Describe the principle of 454 Pyrosequencing. (CO1)
- (b) Write a brief note on JIGP. (CO1)
- (c) What is shotgun method of genome sequencing and how is it different from clone CONTIG strategy ? (CO1)
2. (a) Differentiate between the terms forward and backward genomics. (CO2)
- (b) Write a short note on Gene Editing Technology. (CO2)
- (c) Explain Antisense Technology in detail. (CO2)

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(2)

3. (a) Write short note on comparative genomics and various tools involved in its study. (CO3)
- (b) Differentiate between the following : (CO3)
- (i) Epigenetics and Epigenomics
 - (ii) Toxicogenomics, Pharmacogenomics and Metagenomics
- (c) Comment upon importance of model organisms in omics technology ? Explain *Mus musculus* as model organisms. (CO3)
4. (a) Write short notes on the following : (CO4)
- (i) Role of sucrose and Glycerol in 2D SDS PAGE.
 - (ii) Role of ampholyte molecule in 2D SDS-PAGE .
 - (iii) Role of SDS and PAGE in electrophoresis ?
- (b) Write short notes on the following : (CO4)
- (i) Use of magnetic field in mass spectroscopy.
 - (ii) Detector used in mass spectroscopy.
 - (iii) Application of mass spectroscopy in proteomics.
- (c) Discuss the terms cell free protein synthesis and application of wheat germ extract. (CO4)
5. (a) Illustrate the term transcriptomics and its various applications. (CO5)
- (b) Discuss the working principle of RT-PCR and its application in detection of novel SARS-CoV-2. (CO5)
- (c) Illustrate the fabrication and working of DNA-Microarray and its application in transcriptomics. (CO5)

Mid Semester Examination 2022

Name of the Program: B tech Biotech Semester: VIII

Name of the Course: Entrepreneurship Development

Course Code: TBT802

Time: 1-1/2 Hour Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub questions.
- (ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	Suppose Ms. Malvika wants to open a food joint, which form of business you would recommend to her and why? Support your answer with suitable example.	
OR		CO1
(b)	Aditi started her boutique business near an industrial area and she was not getting clients as she had expected. What are the factors which have been ignored before starting the boutique? Suggest the various factors that she has to look into before venturing into business.	
Q2	(10 marks)	
(a)	Distinguish between creativity, innovation and invention.	
OR		CO1
(b)	Great ideas just happen. Discuss the statement with suitable example.	
Q3	(10 marks)	
(a)	What is Corporate Entrepreneurship and why is it important?	
OR		CO1/CO2
(b)	Discuss the cultural, economic, and political and personality factors that affect entrepreneurial growth.	
Q4	(10 marks)	
(a)	Explain BRIE model of Entrepreneurship process.	
OR		CO2
(b)	"E-Commerce has transformed the globe into a 24x7 marketplace." Discuss.	
Q5	(10 marks)	
(a)	Why IPR is important for Entrepreneur, please justify with example?	
OR		CO2
(b)	What is MSME? Why MSME is considered backbone of Indian Economy?	

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TBT-802

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END SEMESTER EXAMINATION, May, 2022**

ENTREPRENEURSHIP DEVELOPMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) "Logic will get you from A to B. Imagination will take you everywhere." Comment. (CO1)

(b) "The phenomenon of Liberalization, Privatization and Globalization has created a favorable environment of entrepreneurship in India." Critically examine the statement and suggest how the youngsters can make the best use of this. (CO1)

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- (c) Define Rural, Social and Women Entrepreneurship. What is their contribution to Indian Economy ? (CO1)
2. (a) Which form of business ownership in your viewpoint is suitable to undertake for budding entrepreneurs like you ? Justify with example. (CO2)
- (b) Give some reasons why an individual entrepreneur might succeed in bringing a product to the market where the government or a large corporation might fail. (CO2)
- (c) "One of the essential objectives of managing the market is creating demand." Explain. (CO2)
3. (a) What are the traditional sources of Finance ? Describe briefly. (CO3)
- (b) Think honestly of your entrepreneurial intentions. Make a list of problems that you are likely to face by becoming an entrepreneur. (CO3)
- (c) A person has decided to start a smart leather belt manufacturing unit. He/she is not aware of the various formalities involved in the process of setting up the unit. Explain to him/her the procedure. (CO3)
4. (a) Define DC, NDA & ROC, with example. (CO4)
- (b) What is Trademark ? Why business person need to register for Trademark ? Explain with example. (CO4)
- (c) Explain any *three* types of business model with example. (CO4)

(3)

5. (a) What is the importance of Business Model in Developing Business Plan ? (CO5)
- (b) Create Business Model Canvas of Skype and Ola Cabs, both are service based companies. (CO5)
- (c) India will be a nation of entrepreneurs in the 21st century. Explain why and how ? (CO5)